



Digital Egypt Pioneers Initiative - DEPI
Round 3, 2025



CAPSTONE PROJECT

MANUFACTURING DOWNTIME ANALYSIS



A data-driven look into production efficiency and downtime causes

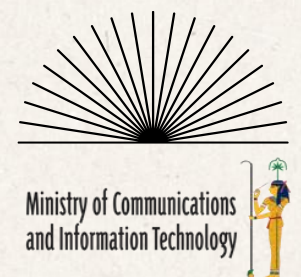
TRACK:
Data Analytics

MAJOR:
Microsoft Power BI Specialist

PRESENTED BY:
CLS-CAI3-DAT2-G5 Team 1

PRESENTED TO:
DEPI

INSTRUCTOR:
Mahmoud Seraj



Team Members



1. Hamza Bahgat
21095905



2. Hussien Habashy
21098002



3. Kareem AbdelHamied
21086585



4. Mohamed Ali
21045195



5. Mohammed Ezzat
21062469



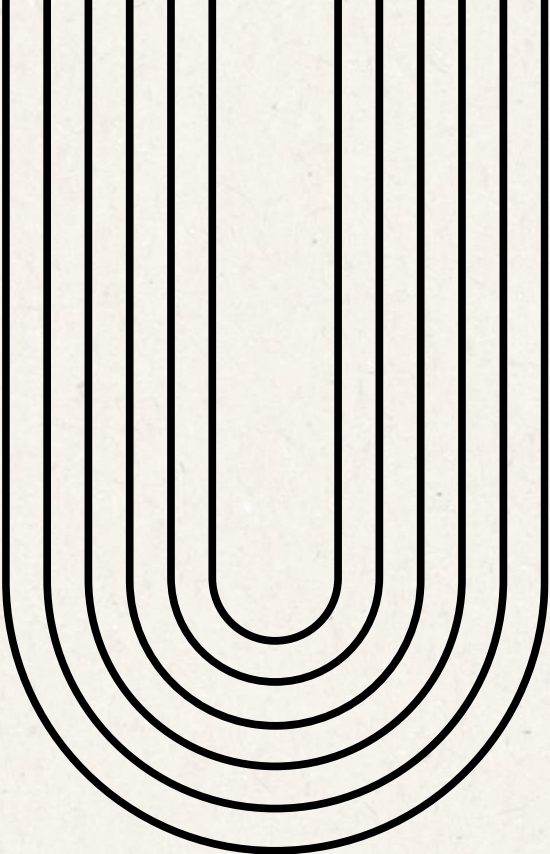
6. Sohaib Atef
21115360

Agenda

02	Project Overview
05	Data Sources & Structure
09	Key Performance Indicators
12	Dashboard Introduction
21	Key Insights
26	Recommendations

Project Overview

This project analyzes manufacturing downtime using **Power BI** to identify key causes and improve production efficiency



OBJECTIVE

Reduce downtime and improve efficiency



SCOPE

Covers multiple products and batches and capturing incidents across operators



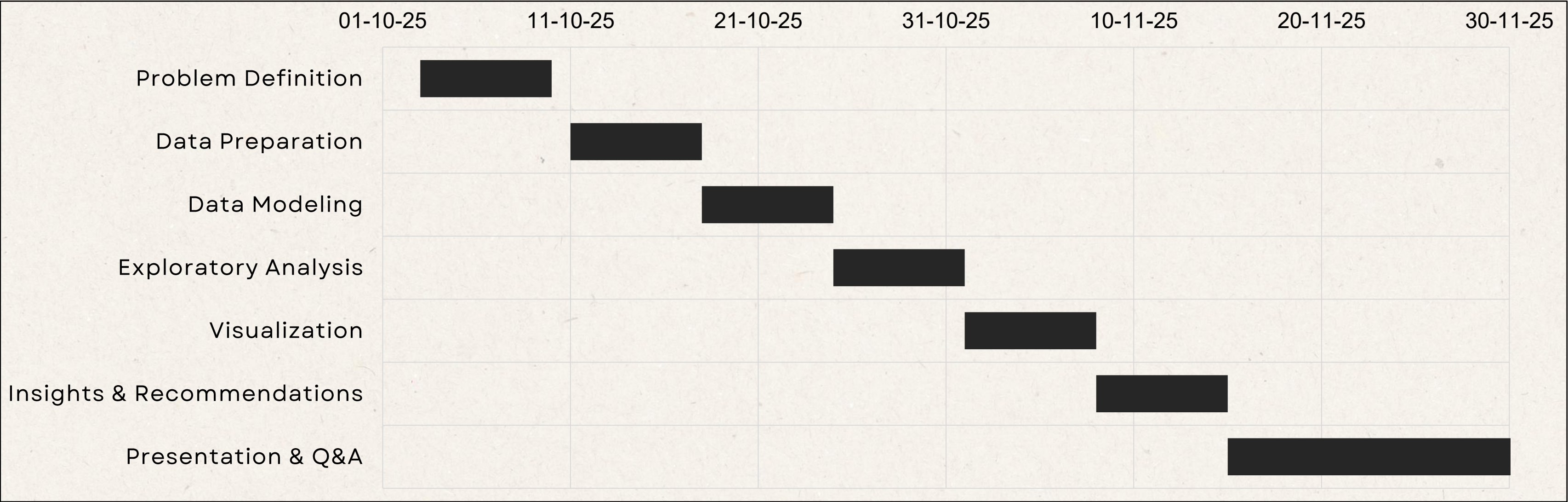
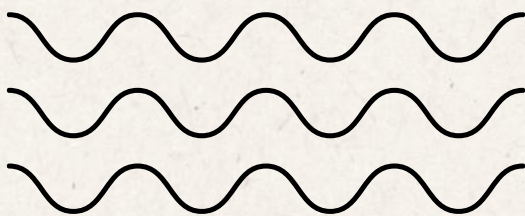
OUTCOME

Identify root causes and suggest improvements

1 Line 6 Products 38 Batches 61 Incidents 4 Operators 5 Days

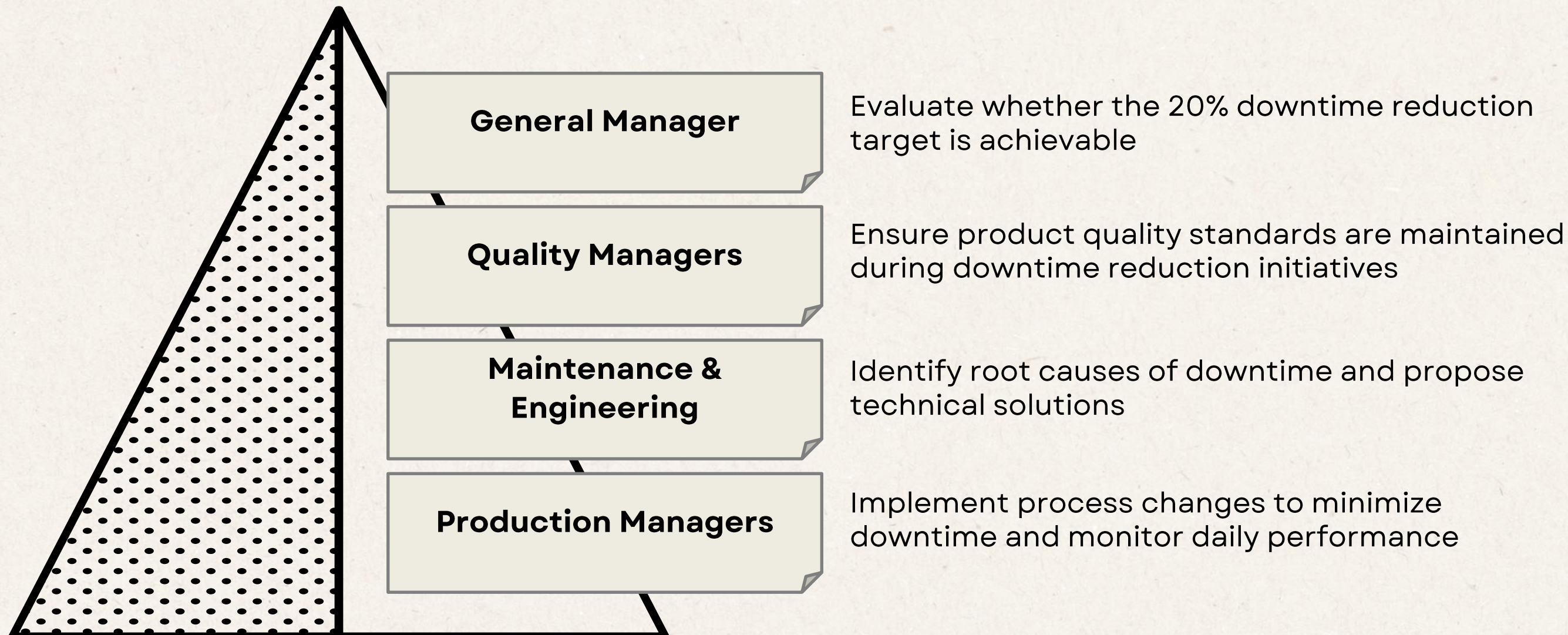
Timeline

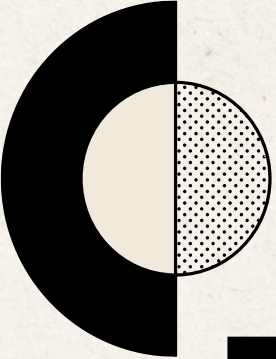
Structured timeline ensured systematic progress toward reducing downtime



Audience

Our findings directly support the needs of key stakeholders across all levels of production



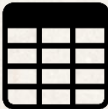


Data Source & Structure

This dataset was provided by the Initiative Management team and includes supporting files such as Excel sheets and metadata for analysis

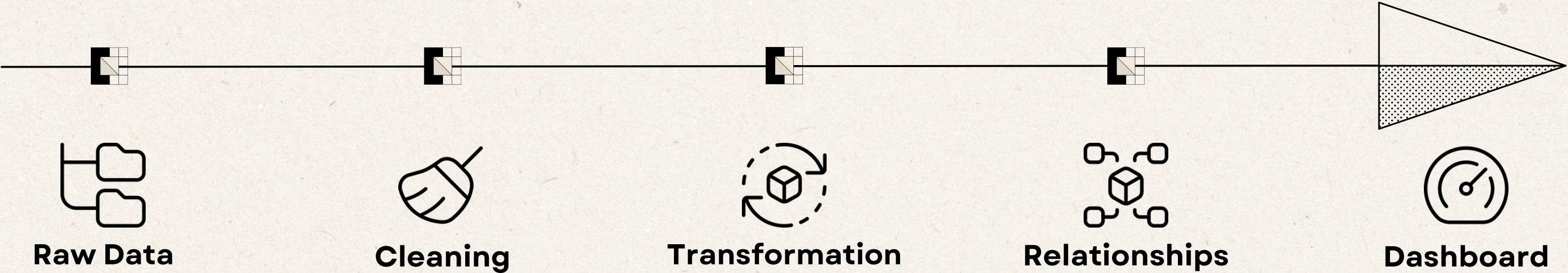
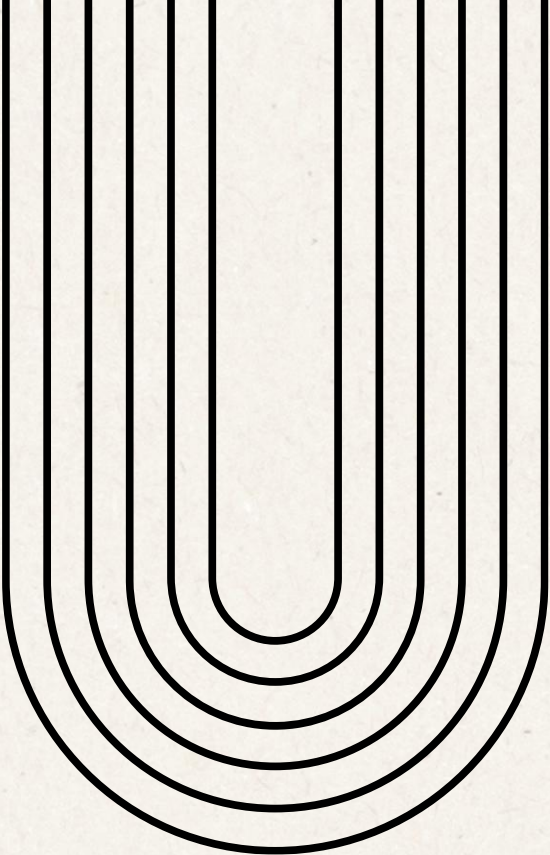


Manufacturing Downtime

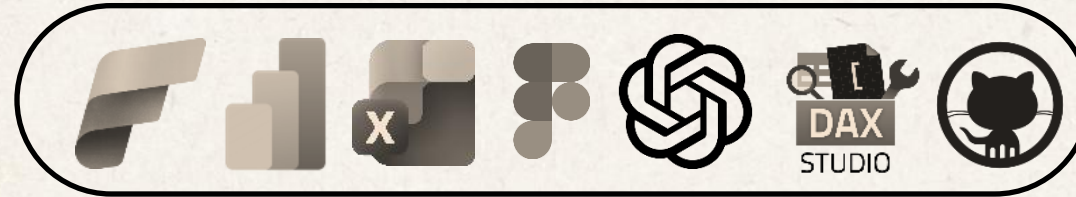
 data_dictionary.csv (Metadata)	 Manufacturing_Line_Productivity.xlsx			
	 Line Productivity	 Products	 Downtime Factors	 Line Downtime

Data Preparation Process

Structured preparation ensured reliable insights for downtime analysis



Tools Used



Analytical Workflow: From Data to Decisions

DATA PREPARATION

- › Excel
- › Power Query



DESIGN & VISUALIZATION

- › Figma
- › Power BI Desktop



MODELLING & ANALYSIS

- › Data Analysis eXpressions – DAX
- › Chat-GPT
- › DAX Studio



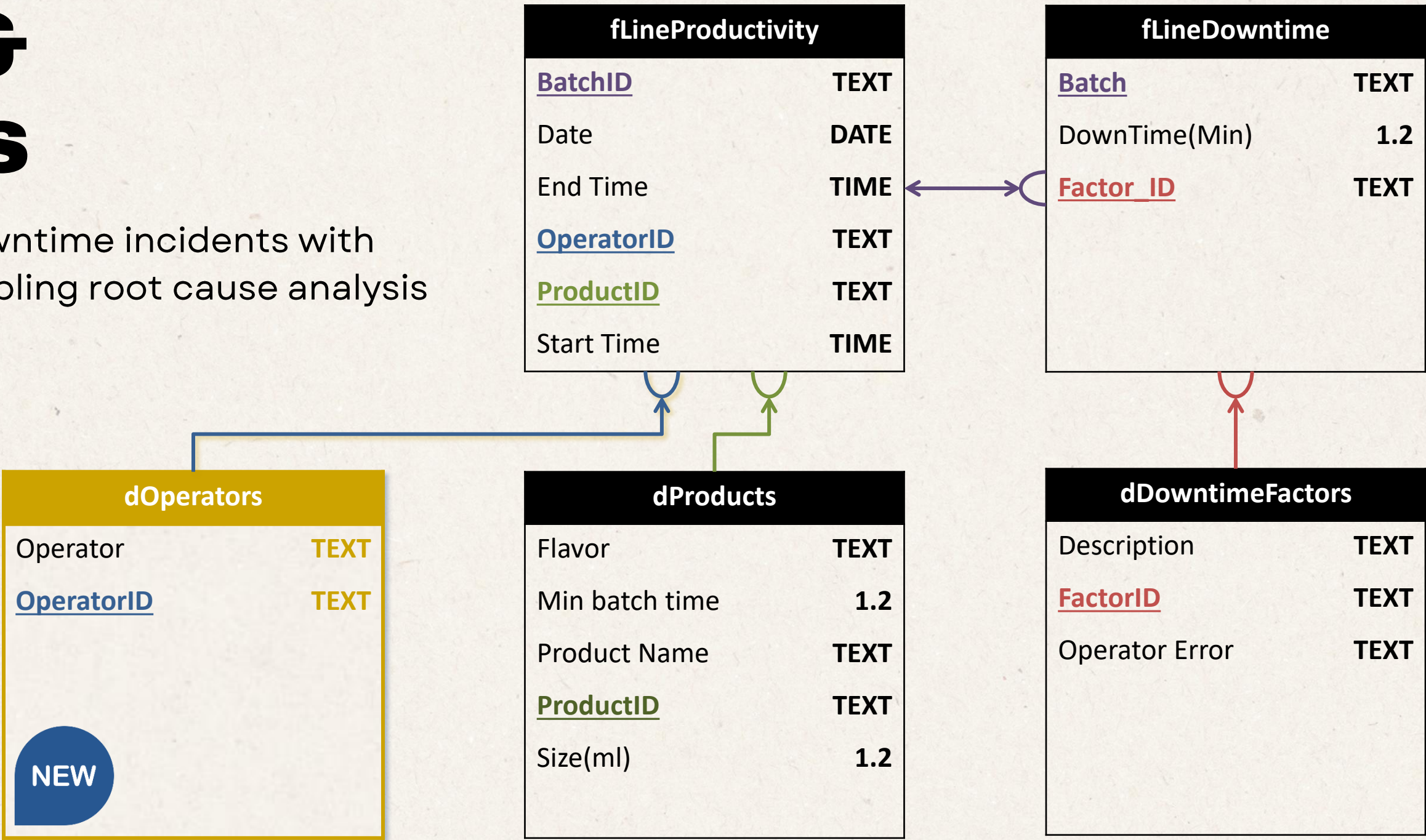
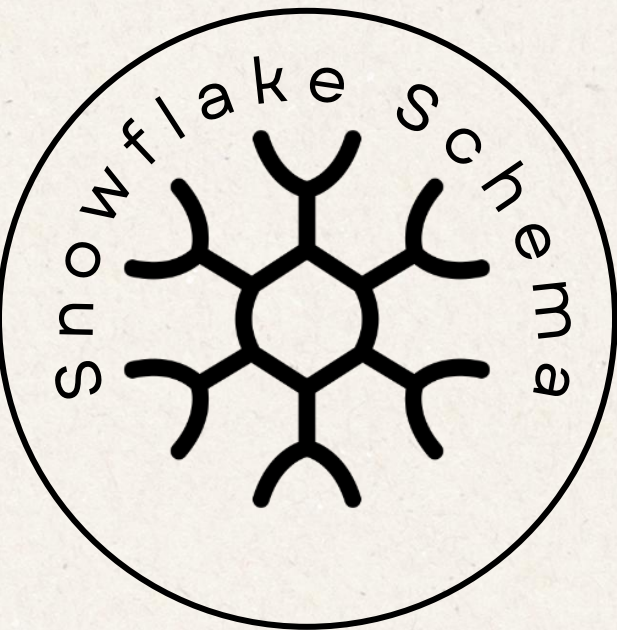
COLLABORATION & SHARING

- › Power BI Service
- › GitHub



Data Model & Relationships

This model allowed us to connect downtime incidents with specific operators and products, enabling root cause analysis



Key Performance Indicators

Our target is to Increase the line availability by **20%**

09/26

Line Availability %

Present: **64%**
Target: **84%**

We need a 20% increase to achieve higher operational efficiency.

Operator Error %

Present: **56%**
Target: **36%**

Reducing operator errors will directly improve productivity

AVG Downtime

Present: **00:23**
Target: **00:18**

Lowering downtime is essential to boost overall output

Data Analysis eXpressions (DAX)

10/26

Line Availability = $\frac{\text{Net Operating Time (Total Production Time - Total Downtime)}}{\text{Total Production Time}}$

DIVIDE([Net Operating Time (M)], [Total Production Time (M)], 0)

DAX

Total Production Time = *End Time - Start Time*

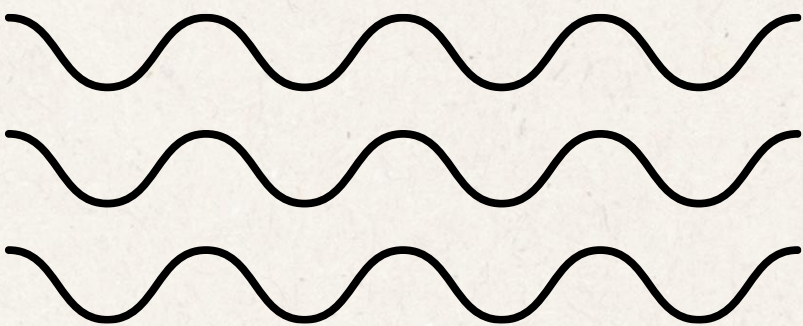
```
SUMX(fLineProductivity,  
    VAR s = fLineProductivity[Start Time]  
    VAR e = fLineProductivity[End Time]  
    VAR eAdj = IF( e < s, e + 1, e )  
    RETURN DATEDIFF(s, eAdj, MINUTE))
```

DAX

Total Downtime =
 $\sum \text{Downtime Minutes}$

```
VAR TotalDownTime =  
SUM(fLineDowntime[DownTime(Min)])  
RETURN  
IF(  
    ISBLANK(TotalDownTime),  
    0,  
    TotalDownTime)
```

DAX



Downtime Operational Dashboard

Interactive visuals to track downtime trends and identify root causes

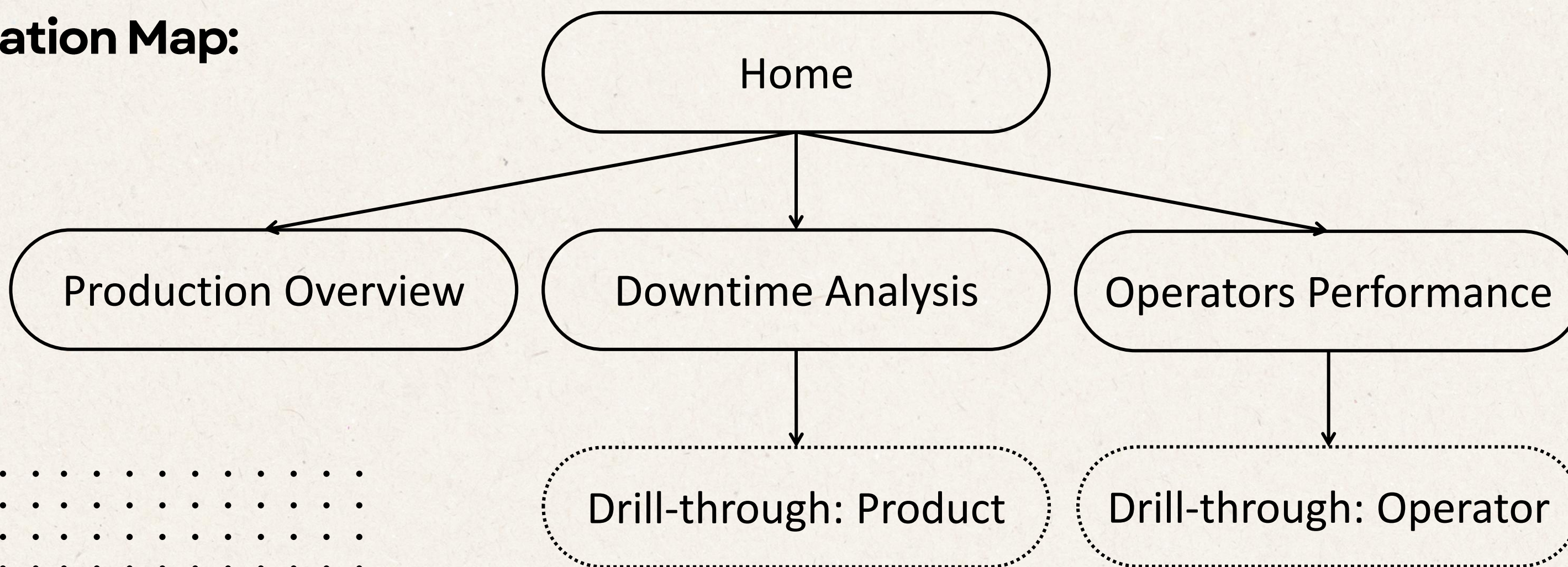


SKILLS SHOWCASED: Figma Design Buttons and Bookmarks Drill-through & Filter Tool Tip Calculated Tables & Measures

Downtime Operational Dashboard

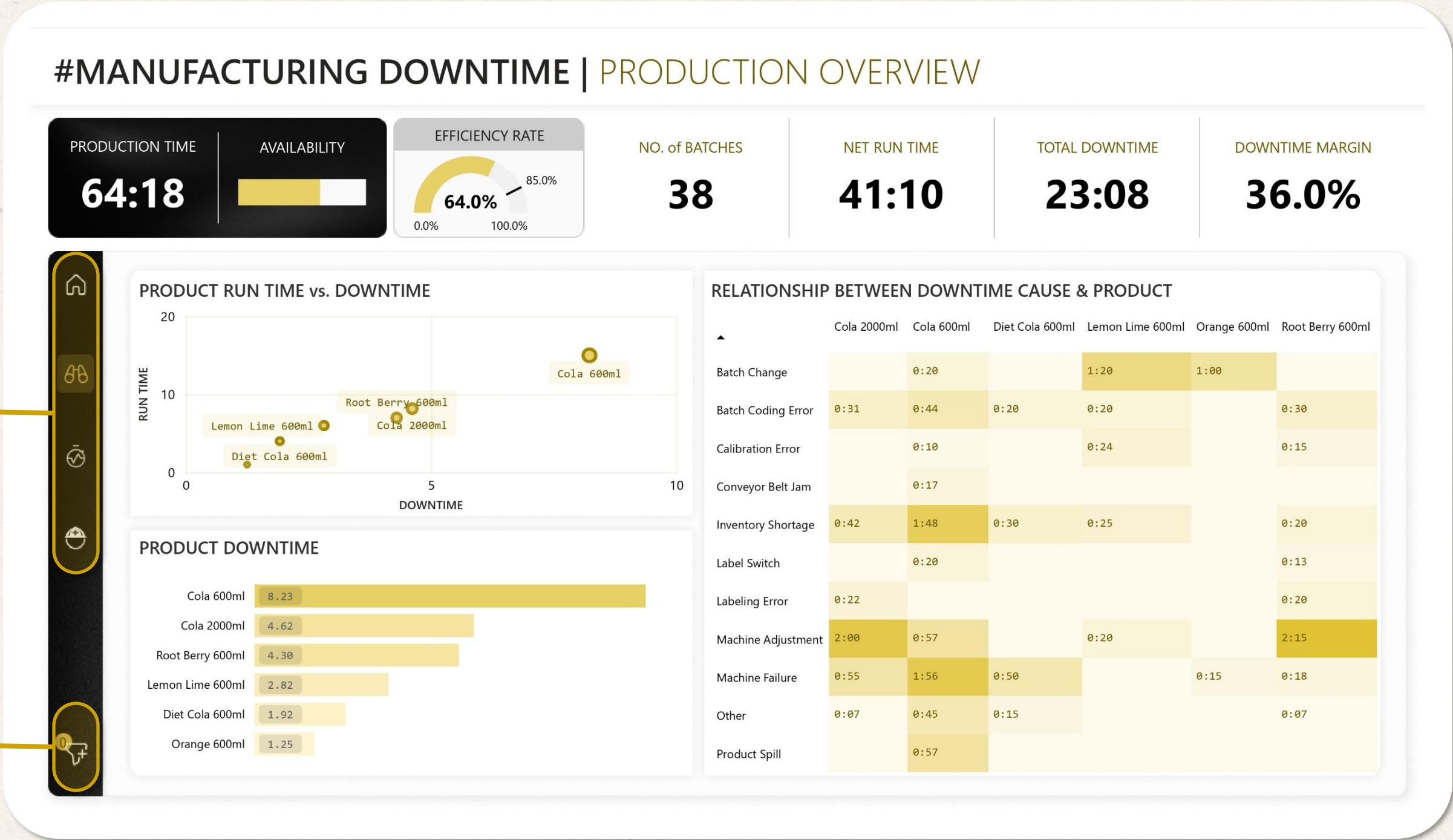
11/26

Wireframe & Navigation Map:



Downtime Operational Dashboard

Page 1:
Production Overview



Main page navigators

Filter button & counter

Downtime Operational Dashboard

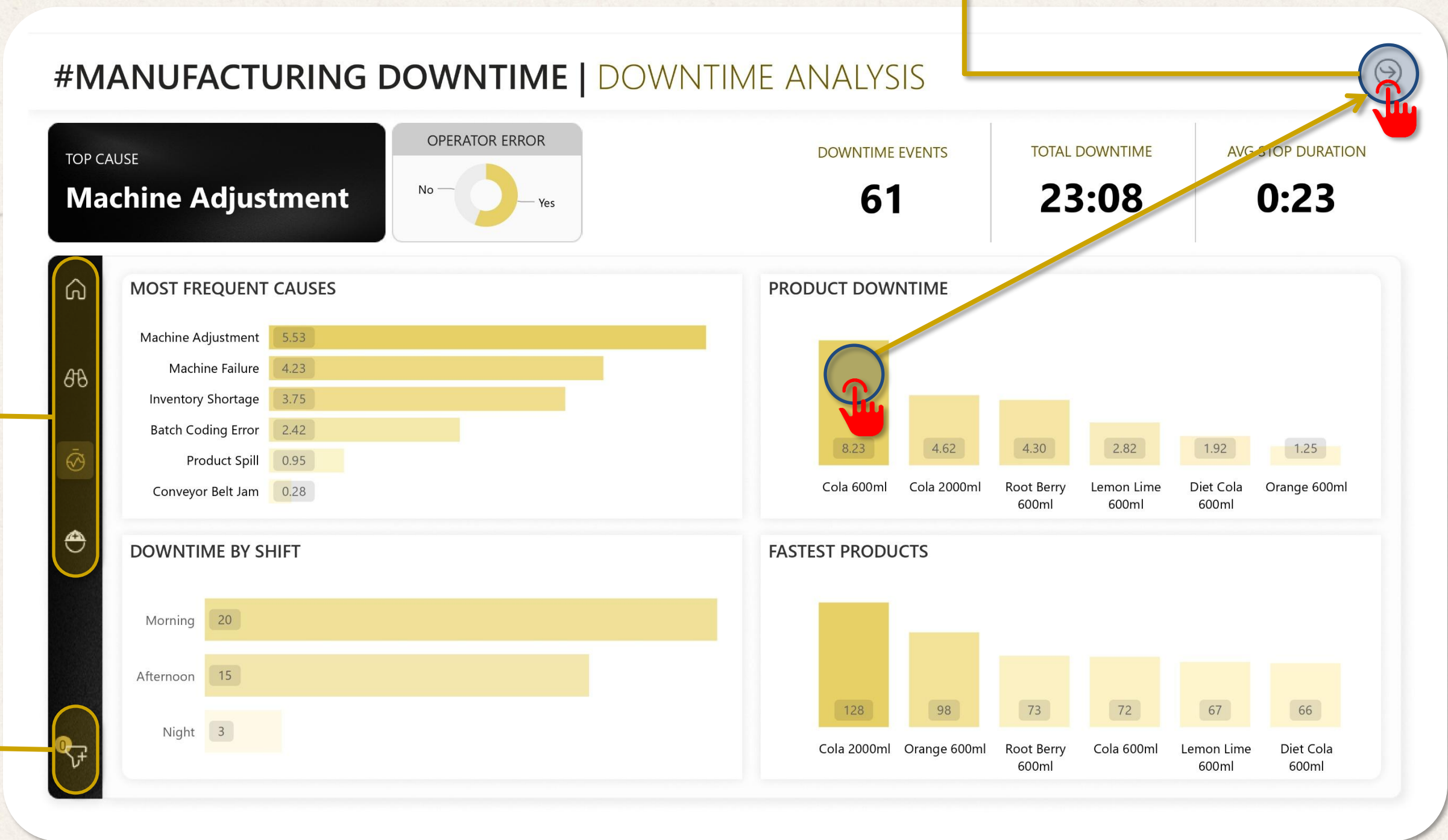
Page 2:
Downtime Analysis

Product drill-through page

14/26

Main page navigators

Filter button & counter

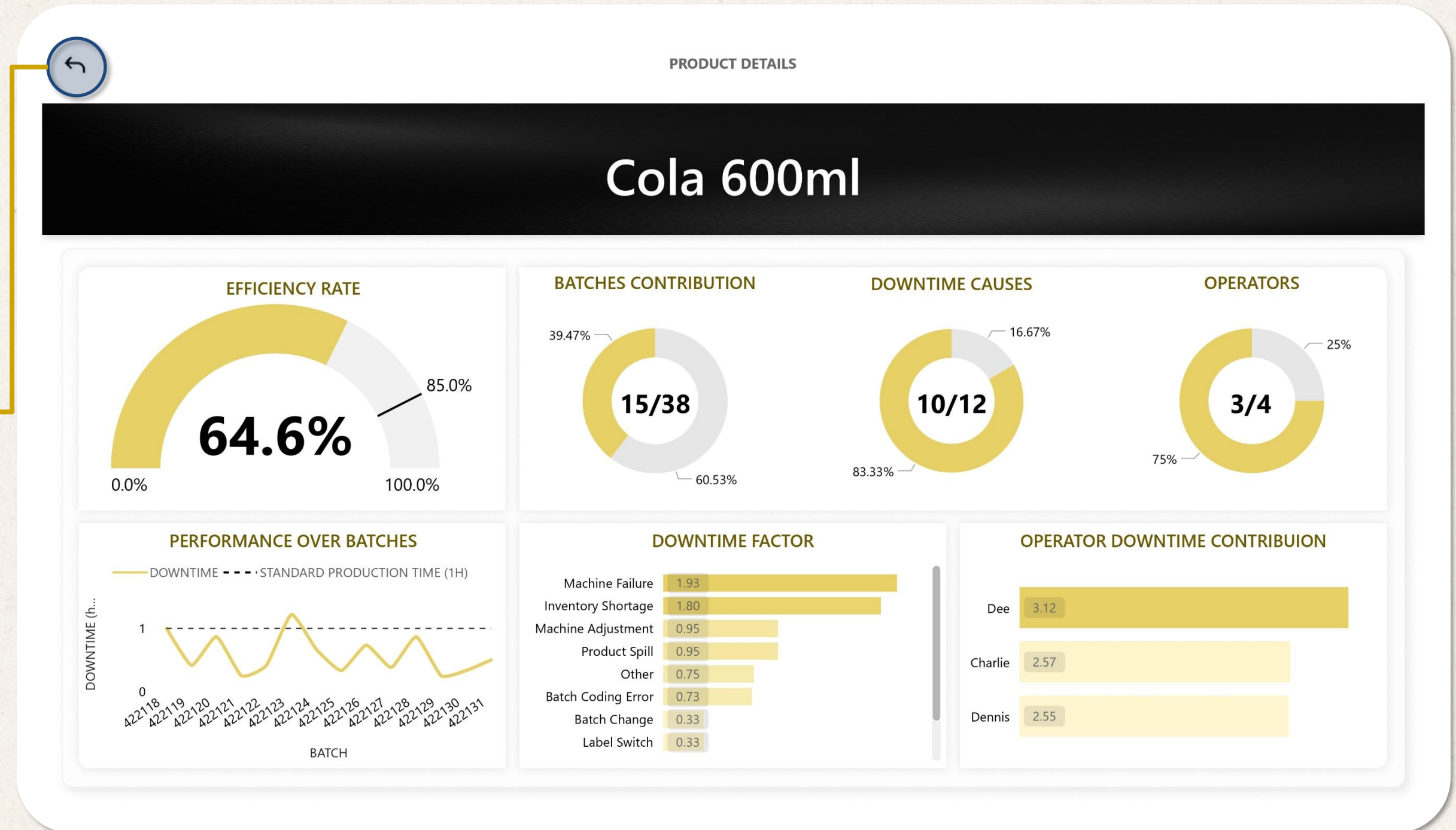


Downtime Operational Dashboard

15/26

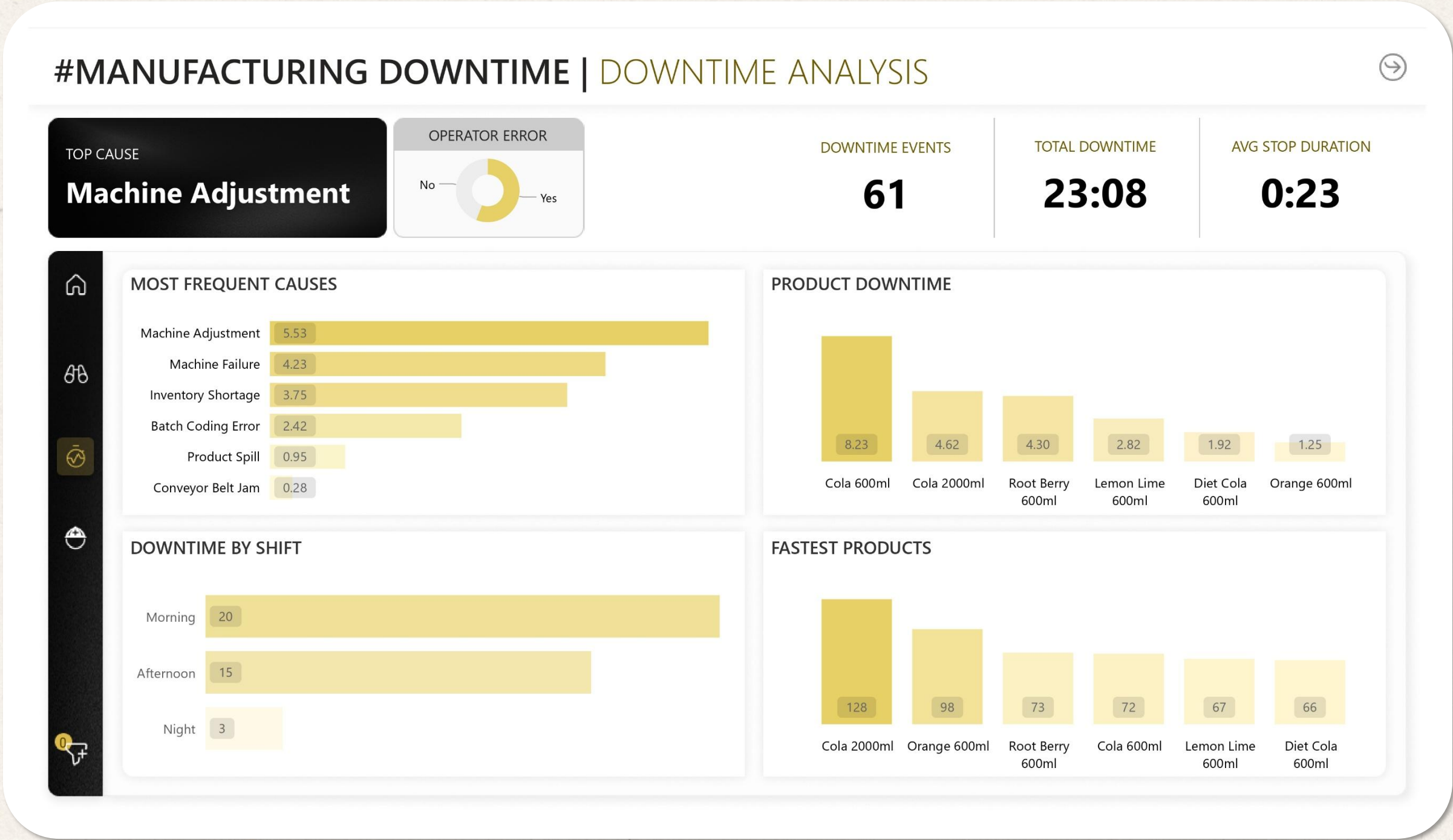
Page 2.1: Product Details

“Downtime Analysis” page



Downtime Operational Dashboard

Page 2: Downtime Analysis



Downtime Operational Dashboard

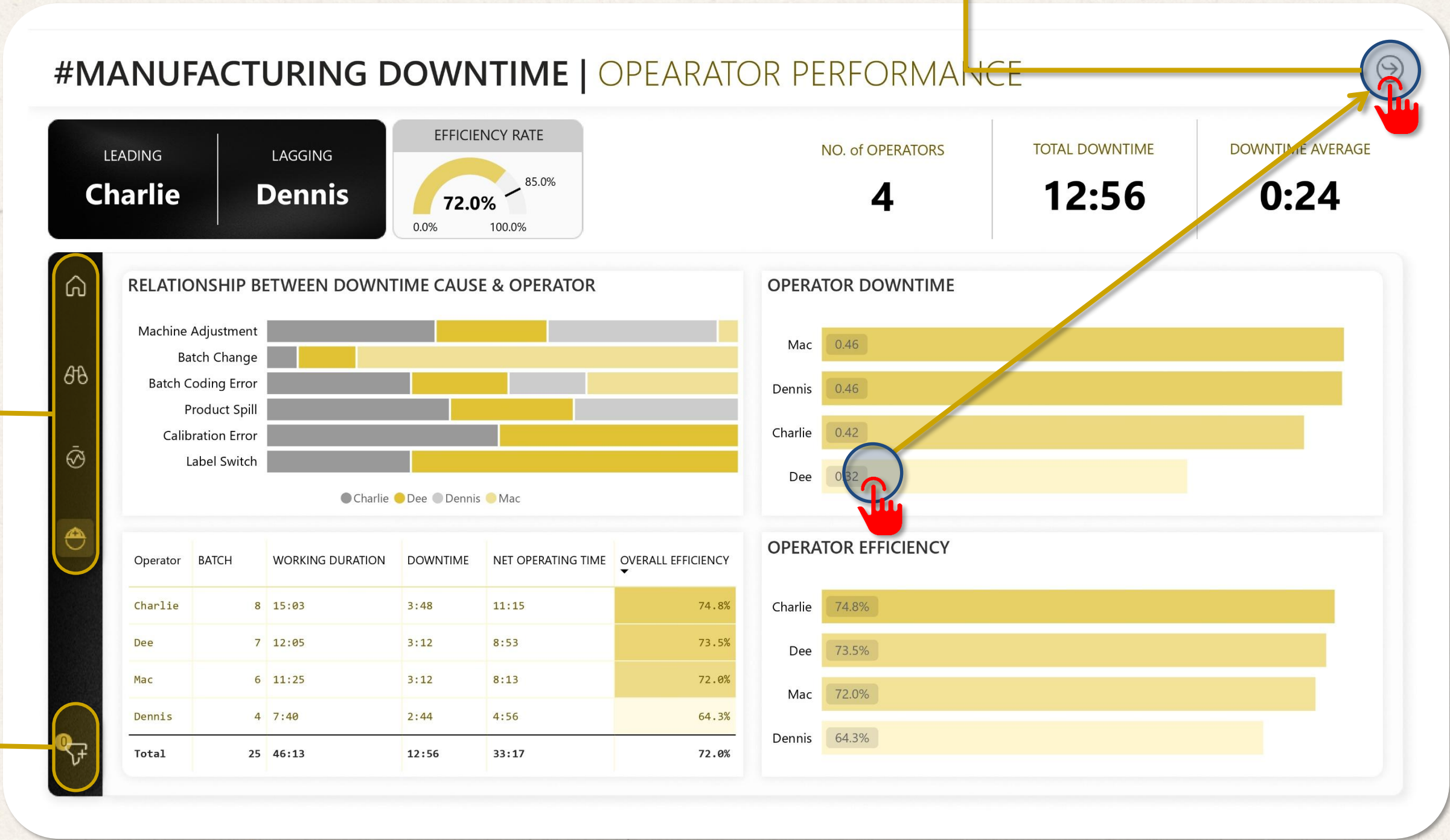
Page 3:
Operator Performance

Operator drill-through page

17/26

Main page navigators

Filter button & counter

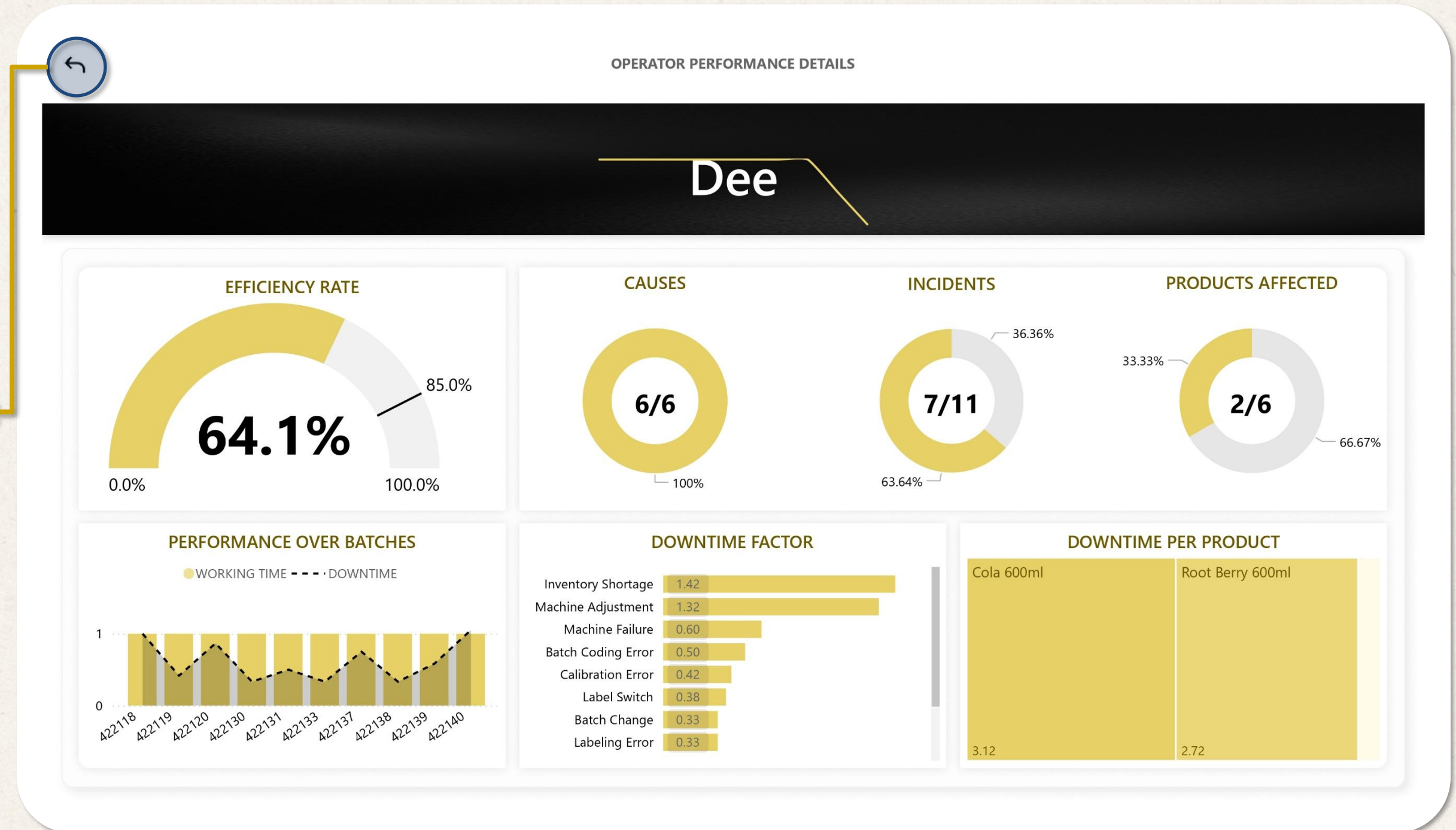


Downtime Operational Dashboard

18/26

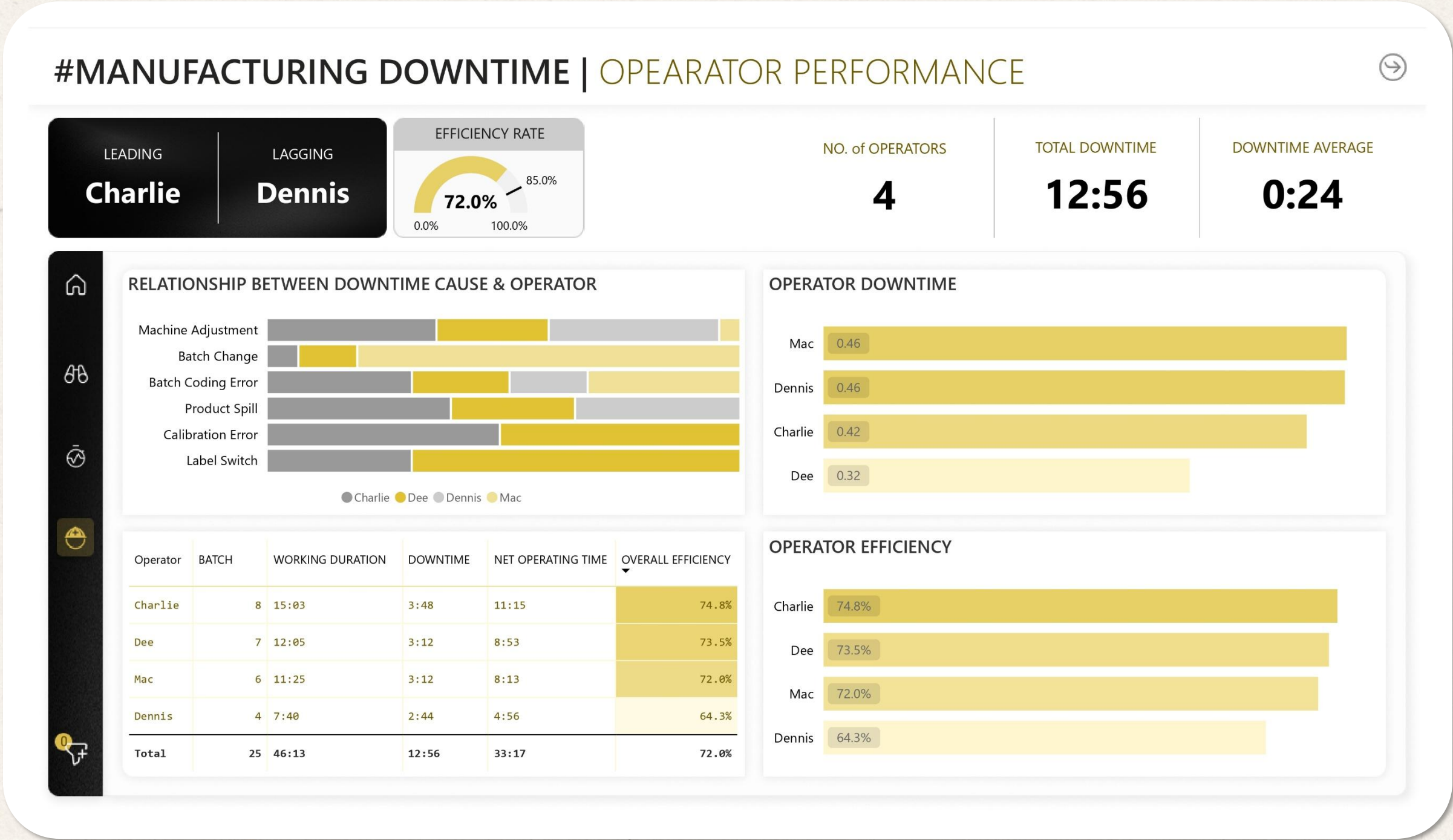
Page 3.1: Operator Details

“Operator Performance” page



Downtime Operational Dashboard

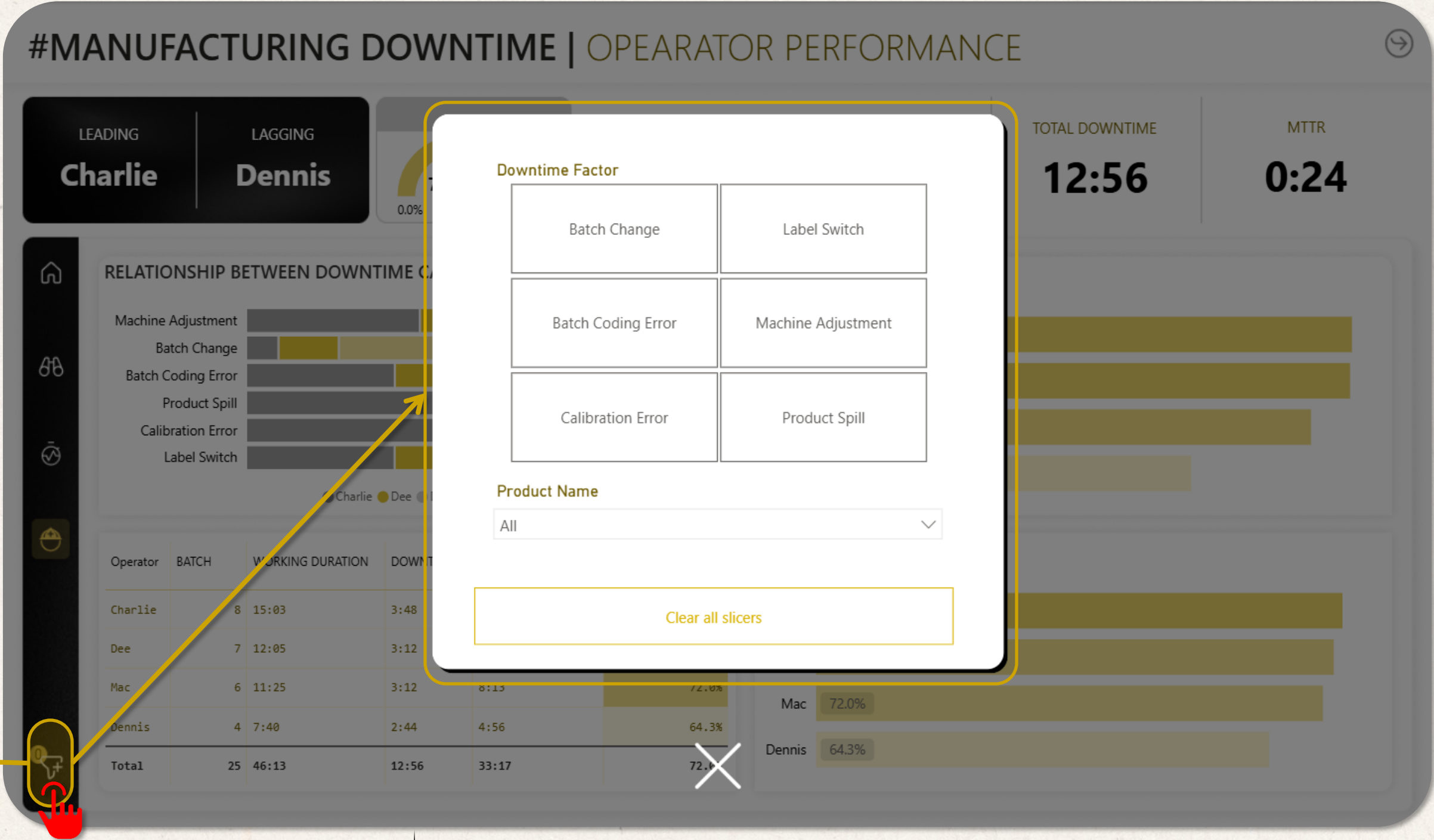
Page 3: Operator Performance



Downtime Operational Dashboard

Page 3: Operators Filter Pop-up

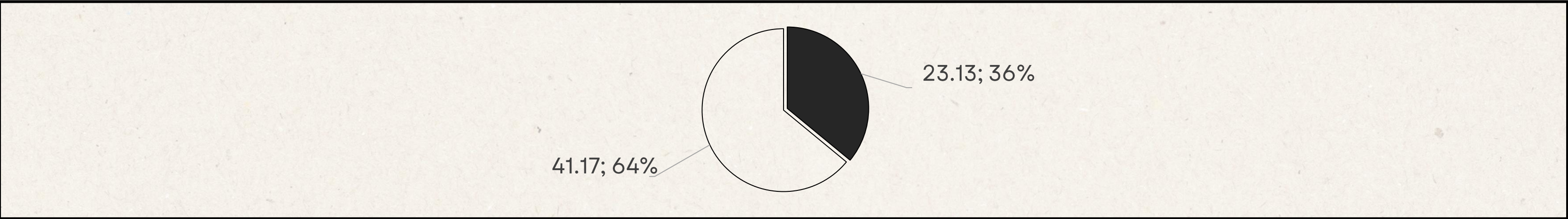
Filter button & counter



Key Insights

Our analysis revealed the main drivers of downtime and efficiency gaps

1. Downtime Is Significantly Above Normal Operational Limits



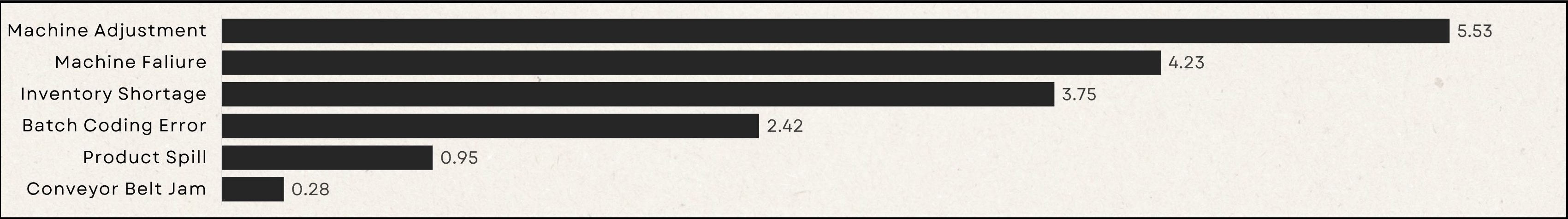
- › Total downtime recorded was 36%, limiting line availability to 64%!

Why It Matters?

- › Reduced availability puts pressure on delivery schedules and reduces total output capacity.

Key Insights

2. Machine-related Issues Account For Most Non-operator Downtime, Led By ‘Machine Adjustment’



- › “Machine Adjustment” alone causes almost a quarter of total losses.
- › “Machine Failure” and “Inventory Shortage” together add over 8 hours of extra downtime.

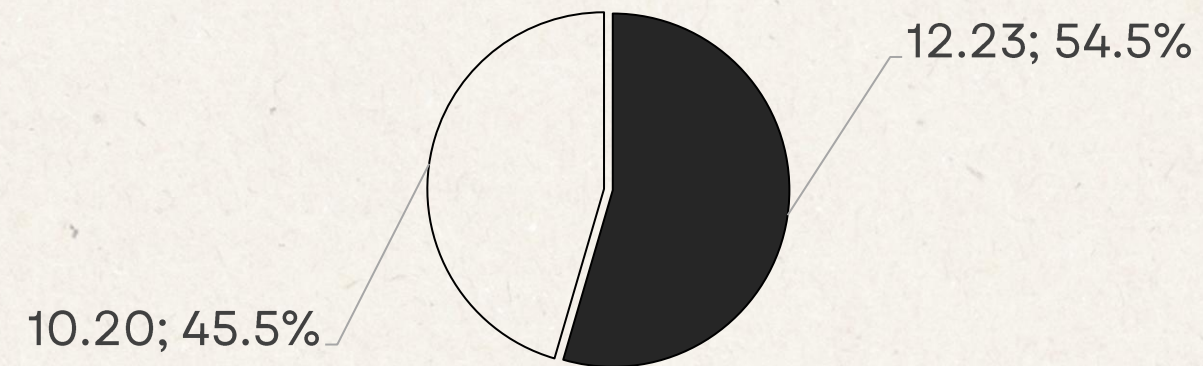
Why It Matters?

- › Recurring mechanical problems slow production, reduce output, and increase maintenance workload.

Key Insights

23/26

3. Operator Errors Are the Main Driver



- › 54% Operator Error – 46% Other Causes.
- › More than half of all incidents came directly from operator mistakes.

Why It Matters?

- › This makes operator performance the most critical factor affecting overall downtime.

Key Insights

24/26

4. Some Operators Consistently Contribute More Downtime Than Others



- › Downtime per operator varies, with Mac & Dennis at the highest levels

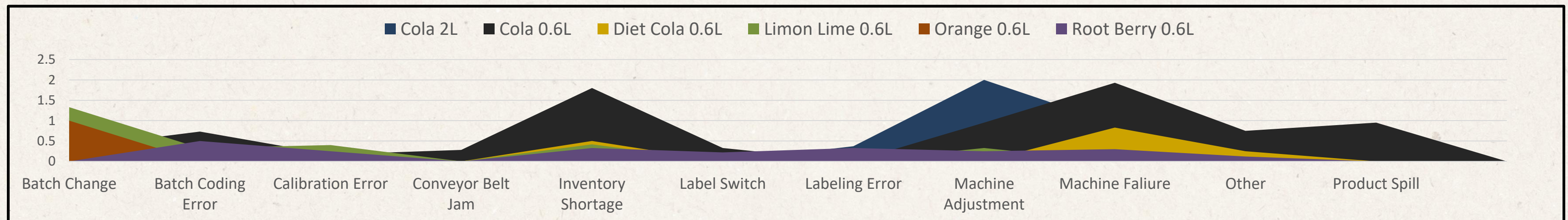
Why It Matters?

- › Targeted coaching can reduce error rates and stabilize performance across the team.

Key Insights

25/26

5. Certain Downtime Causes Are Concentrated on Specific Products



- › There is a strong relationship between the “Cola 2L” and the “Machine Adjustment”.
- › “Cola 600ml” -in particular- appears across 7 downtime factors.

Why It Matters?

- › This suggests that targeted troubleshooting for these products can significantly reduce overall downtime.

Recommendations

26/26



Operator Performance

- › Conduct targeted retraining for Mac and Dennis.
- › Introduce Standard Operating Procedures (SOPs).



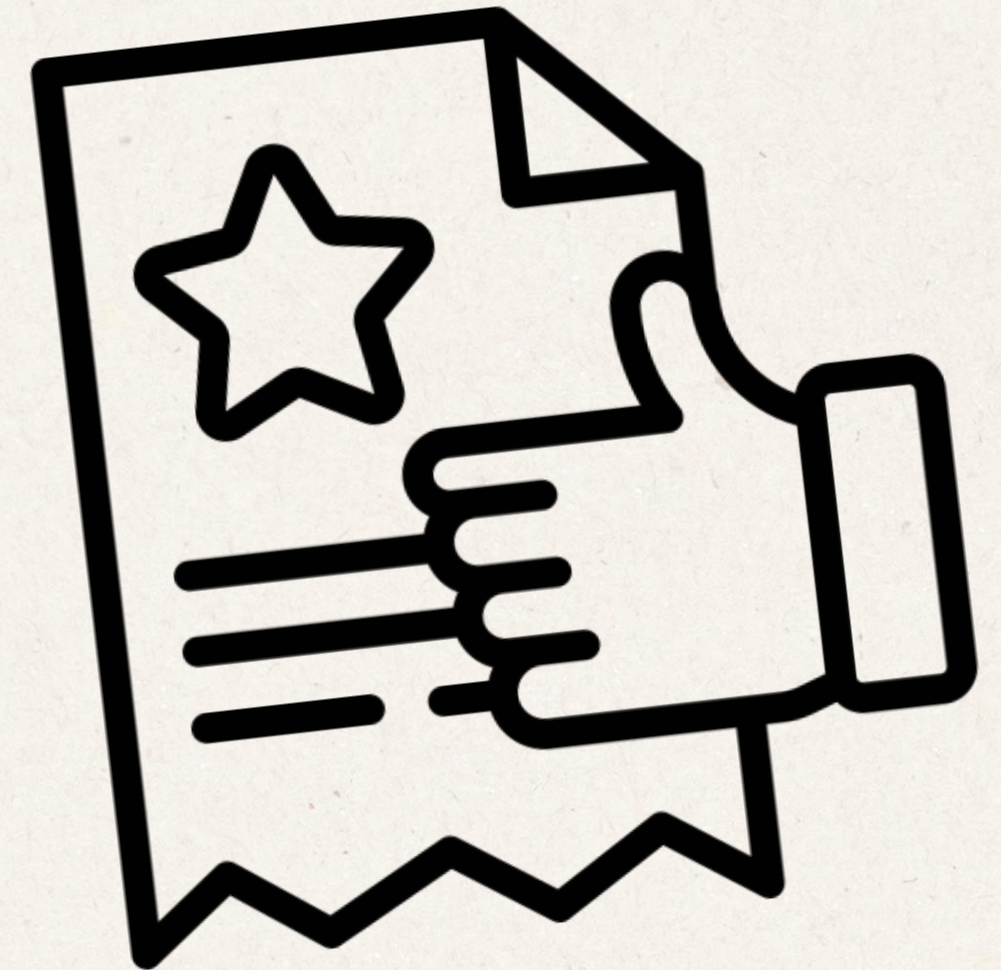
Equipment & Maintenance

- › Perform full diagnostic inspection of machines associated with high downtime.
- › Implement a structured Preventive Maintenance (PM) plan.



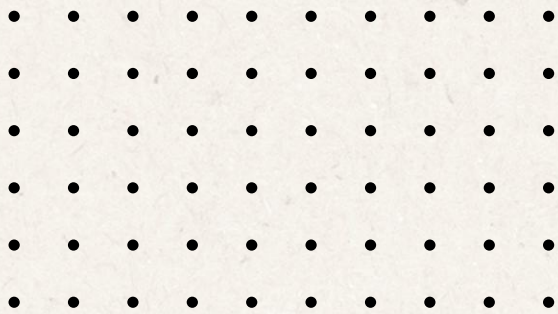
Product Specific

- › Create dedicated setup and calibration checklists for Cola 2L.
- › Perform test verification runs after each product changeover.





DEC 2025



THANK YOU

Visit the GitHub Repo



Please get in touch if you have any questions
or concerns about this report.